

Special39 / competency practice — Electricity Paper 03 — Energy Budgets, Joule Heat and Practical Decisions

CLASS 10 • ELECTRICITY • ORIGINAL PRACTICE

Original Class 10 Electricity • 39 questions • 80 marks • competency practice

Original chapter practice — not an official examination paper, a prediction, or an official chapter weightage.

39 questions • 80 marks • 180 minutes

Original Class 10 Electricity chapter practice: 39 compulsory questions, broad chapter coverage and evidence-led decisions. Not an official paper, chapter-weightage claim or prediction.

Paper structure

Section A | Questions 1–20 | $20 \times 1 = 20$ marks

Section B | Questions 21–26 | $6 \times 2 = 12$ marks

Section C | Questions 27–33 | $7 \times 3 = 21$ marks

Section D | Questions 34–36 | $3 \times 5 = 15$ marks

Section E | Questions 37–39 | $3 \times 4 = 12$ marks

Instructions

Attempt all questions. There are no internal OR choices. Sections A–E form the question paper. Detailed solutions and marking guidance begin only after Question 39; source-pattern and competency rationale follow in a separate appendix.

General instructions

All 39 questions are compulsory. No internal choices. Time: 180 minutes. Maximum marks: 80.

Section A: Q1–20, 1 mark each. Section B: Q21–26, 2 marks each. Section C: Q27–33, 3 marks each. Section D: Q34–36, 5 marks each. Section E: Q37–39, 4 marks each. Subpart marks are stated in each question.

Unless explicitly stated otherwise, voltage sources and switches are ideal, connecting wires have negligible resistance, ammeters have zero resistance and voltmeters have infinite resistance. Resistances remain constant at the stated operating values; comparisons of resistivity use the same temperature. Treat steady stages separately.

For purely resistive devices all electrical input energy becomes heat in the resistors; do not assume all of that heat reaches a useful load unless stated. Any useful-transfer fractions or other losses are stated explicitly. Device ratings apply at rated voltage; at a changed voltage

use the stated fixed-resistance model.

Use $I = Q/t$, $V = W/Q$, $V = IR$, $R = \rho L/A$, $H = I^2 R t$ and $P = VI$ as appropriate. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$; $1 \text{ Wh} = 3600 \text{ J}$. No calculus or specific-heat formula is required. For a straight-line power ramp, energy equals average endpoint power multiplied by elapsed time (equivalently, use rectangles and triangles).

Figures are original and form part of the questions. Use numerical labels, not drawing scale, for dimensions. Show working and units for written answers.

Shared options for Q17-20

A: both true and R correctly explains A

B: both true but R does not correctly explain A

C: A true but R false

D: A false but R true

Section A

Question 1 | 1 mark

special-39-03-q01

A net 5.0×10^{18} electrons pass a cross-section of a wire in one direction during 4.0 s. The magnitude of the charge on each electron is $e = 1.6 \times 10^{-19}$ C. What is the magnitude of the average current? [1 mark]

- A. 0.80 A
- B. 0.20 A
- C. 3.2 A
- D. 1.25 A

Question 2 | 1 mark

special-39-03-q02

Which statement gives the SI definition of a potential difference of one volt between two points? [1 mark]

- A. One coulomb of charge passes a cross-section each second.
- B. One joule of energy is transferred each second.
- C. One joule of work is done for each ampere of current.
- D. One joule of work is done in moving one coulomb of charge between the points.

Question 3 | 1 mark

special-39-03-q03

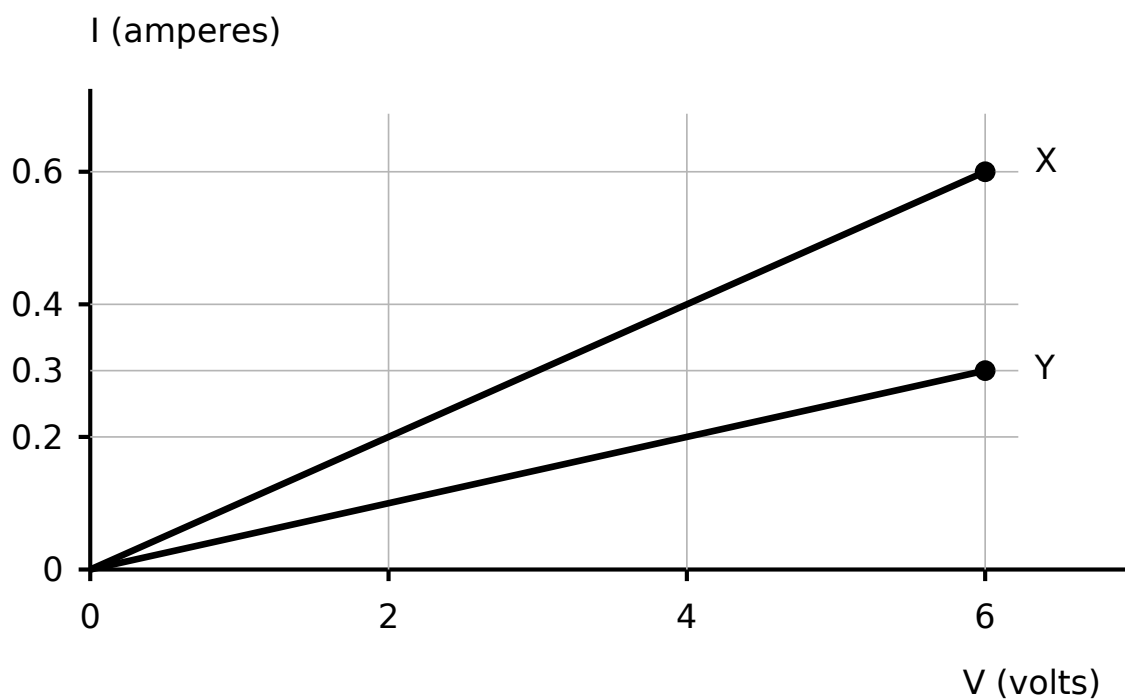
During stage X, 90 C passes in 30 s; during stage Y, 120 C passes in 60 s. The current is steady within each stage. Which statement is correct? [1 mark]

- A. X has 1.5 times the current of Y
- B. Y has twice the current of X
- C. The currents are equal
- D. Y has $\frac{4}{3}$ times the current of X

Question 4 | 1 mark

special-39-03-q04

The figure gives I against V for resistive devices X and Y. Each must operate continuously at 6 V for 10 min. A source can supply at most 1500 J for that interval. Which choice is possible? Read the graph; do not assume that the steeper line means greater resistance. [1 mark]



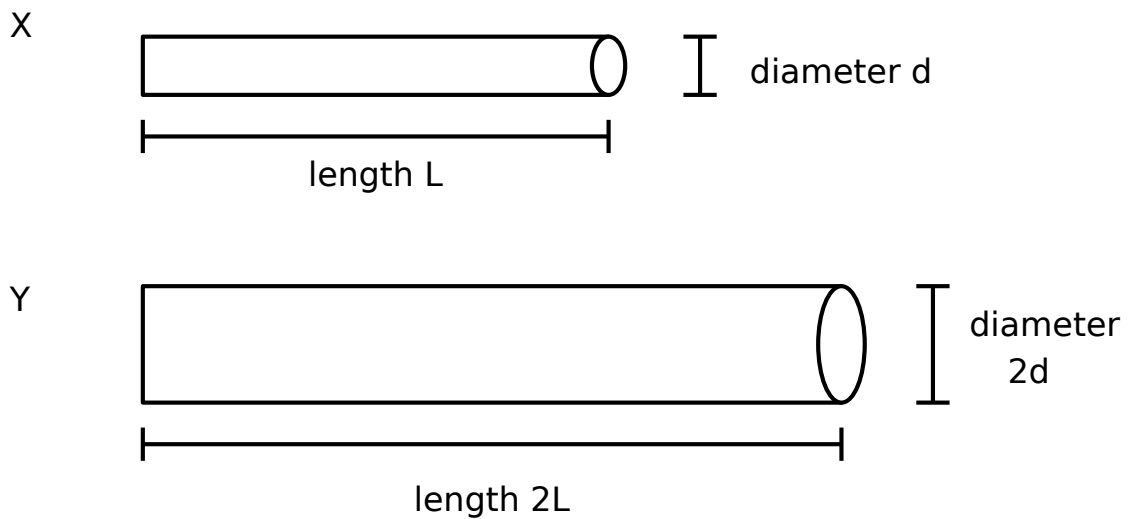
- A. Only X
- B. Both X and Y
- C. Only Y
- D. Neither X nor Y

Question 5 | 1 mark

special-39-03-q05

The figure shows two uniform wires of the same material and temperature; dimensions, not drawing scale, are authoritative. They carry the same steady current for the same time in separate tests. Which prediction is correct? [1 mark]

Same material and temperature; dimensions are labelled.



Not to scale.

- A. Y produces twice the heat of X
- B. Y produces half the heat of X
- C. Y produces the same heat as X
- D. Y produces one-quarter the heat and has lower resistivity

Question 6 | 1 mark

special-39-03-q06

Two controlled drying trials produce the same accepted result. Trial A draws 800 W for 9 min. Trial B draws 500 W for 12 min, followed by 20 W for 15 min. All stated powers are actual constant input powers during their intervals. Which trial uses less electrical energy? [1 mark]

- A. A, by 15 Wh
- B. A, by 20 Wh
- C. B, by 20 Wh
- D. B, by 15 Wh

Question 7 | 1 mark

special-39-03-q07

A fixed-resistance warming tile is rated 60 W at 12 V. At 6 V it must supply 900 J of heat, all of which is useful. Which time is required? [1 mark]

- A. 60 s
- B. 30 s
- C. 15 s

D. 120 s

Question 8 | 1 mark

special-39-03-q08

A constant-temperature resistor gives the following readings: (V in volts, I in amperes) = (1.5, 0.15), (3.0, 0.30), (4.5, 0.90), (6.0, 0.60). Exactly one current was copied incorrectly. Which corrected entry restores consistency with the other three pairs? [1 mark]

A. At 1.5 V, change 0.15 A to 0.30 A

B. At 3.0 V, change 0.30 A to 0.60 A

C. At 4.5 V, change 0.90 A to 0.45 A

D. At 6.0 V, change 0.60 A to 1.20 A

Question 9 | 1 mark

special-39-03-q09

A supplier lists the electrical resistivity of a wire material at 20 °C. An engineer wants to use that value for a heating element operating at a much higher temperature. Which statement is correct? [1 mark]

A. The listed value must apply at every temperature because the material name is unchanged.

B. The operating-temperature resistivity is needed; resistivity can vary with temperature.

C. The resistivity must become zero whenever the element is hot.

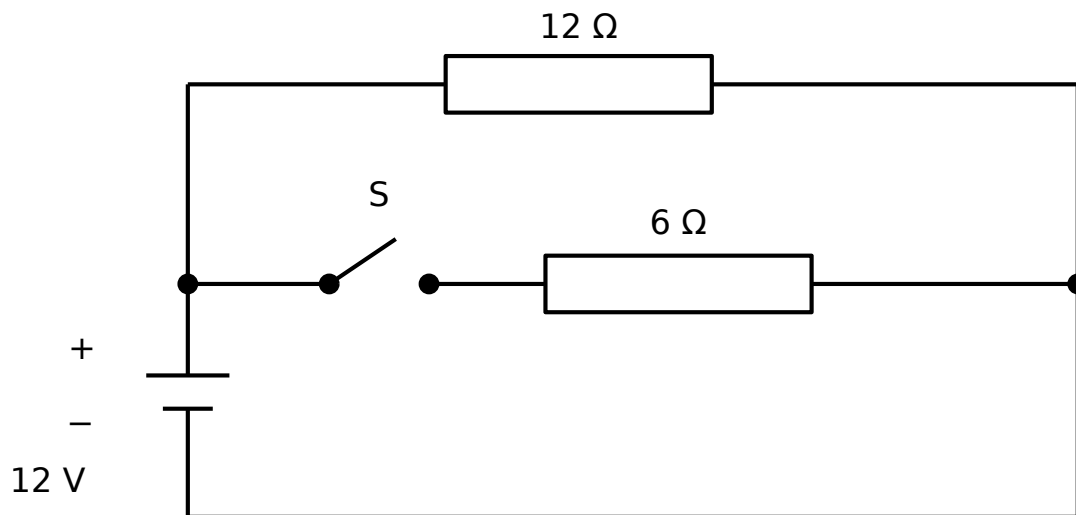
D. The hotter resistivity can be found by dividing the listed value by the wire length.

Question 10 | 1 mark

special-39-03-q10

In the supplied circuit, switch S is initially open. It is then closed while the ideal source stays at 12 V. Which change occurs? All resistances stay constant. [1 mark]

S is initially open



- A. Both branch currents become 1 A
- B. The current in the $12\ \Omega$ resistor becomes 3 A
- C. The source current stays at 1 A
- D. The source current becomes 3 A; the $12\ \Omega$ current stays 1 A

Question 11 | 1 mark

special-39-03-q11

For a fixed resistor, a second trial uses half the first trial's current but lasts twice as long. What is the ratio $H_{\text{second}}/H_{\text{first}}$ of heat generated? [1 mark]

- A. $1/2$
- B. 1
- C. 2
- D. $1/4$

Question 12 | 1 mark

special-39-03-q12

Two identical fixed resistors are connected in series across an ideal voltage source for time t . They are then connected in parallel across the same source. For how long must the parallel arrangement run to use the same total electrical energy? [1 mark]

- A. $4t$
- B. $2t$
- C. $t/4$

D. $t/2$

Question 13 | 1 mark

special-39-03-q13

A steady 2 A is measured immediately before and immediately after a resistor. The resistor transfers 10 J of electrical energy each second as heat. Which interpretation is consistent with both observations? [1 mark]

- A. The resistor consumes 10 C each second
- B. Charge flow is unchanged; 5 J is transferred per coulomb
- C. The exiting current must actually be zero
- D. Equal currents imply zero potential difference

Question 14 | 1 mark

special-39-03-q14

A voltmeter displays +0.4 V when its two leads touch. Across a resistor it displays 2.4 V while an accurate ammeter reads 0.50 A. Assuming this is a constant additive zero error, what should a student do? [1 mark]

- A. Add 0.4 V and report 5.6 Ω
- B. Ignore the offset and report 4.8 Ω
- C. Subtract 0.4 A and report 24 Ω
- D. Subtract 0.4 V and report 4.0 Ω

Question 15 | 1 mark

special-39-03-q15

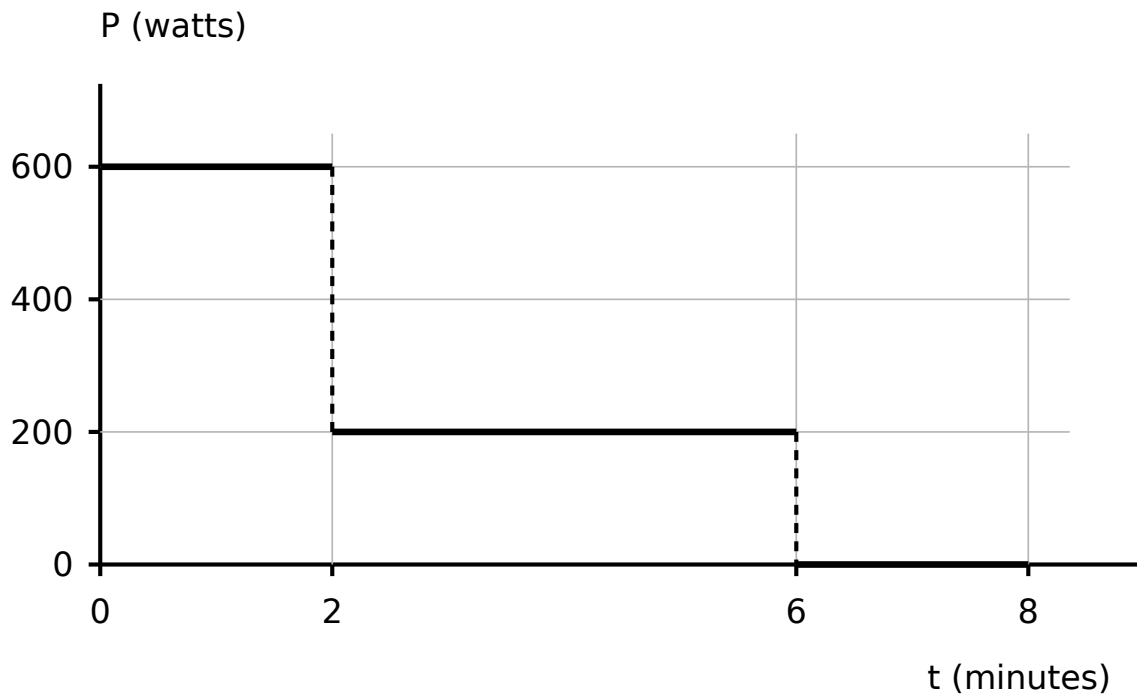
Which pair of properties explains why suitable alloys are commonly used for heating elements? [1 mark]

- A. Relatively high resistivity and resistance to oxidation at high temperature
- B. Zero resistivity and rapid oxidation
- C. Very low melting point and rapid oxidation
- D. Zero resistance and negligible heat production

Question 16 | 1 mark

special-39-03-q16

The graph gives the complete power schedule of a device over 8 min. An alternative draws a steady 300 W for all 8 min. Which comparison is correct? Use rectangular energy blocks, without calculus. [1 mark]



- A. The graph uses 40 Wh
- B. The graph uses 60 Wh
- C. The graph uses $6\frac{2}{3}$ Wh less than the alternative
- D. The graph uses $6\frac{2}{3}$ Wh more than the alternative

Question 17 | 1 mark

special-39-03-q17

Assertion (A): If the resistance of an element is doubled while its voltage and operating time are kept fixed, it produces twice as much heat.

Reason (R): At fixed voltage, the heat produced in time t is $H = V^2t/R$.

Choose the appropriate shared assertion–reason option. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 18 | 1 mark

special-39-03-q18

Assertion (A): A kilowatt-hour is a unit of energy, not power.

Reason (R): An ideal ammeter is connected in series with the component whose current it measures.

Choose the appropriate shared assertion–reason option. [1 mark]

- A. both true and R correctly explains A

- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 19 | 1 mark

special-39-03-q19

Assertion (A): In a steady series circuit, a heater does not make the current measured after it smaller than the current measured before it.

Reason (R): Charge does not accumulate in the heater during steady operation, so equal charge enters and leaves in each time interval.

Choose the appropriate shared assertion–reason option. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Question 20 | 1 mark

special-39-03-q20

Assertion (A): A fixed-resistance heater operated at half its rated voltage for twice a reference time transfers half the energy it would transfer at rated voltage for that reference time.

Reason (R): Halving the applied voltage halves the resistance of a fixed-resistance heater.

Choose the appropriate shared assertion–reason option. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

Section B

Question 21 | 2 marks

special-39-03-q21

A controller supplies 0.40 A for 30 s and then 0.10 A for 90 s. (i) Find the total charge delivered. [1 mark] (ii) A student reports the average current as $(0.40 + 0.10)/2 = 0.25$ A. Test this report using the total charge and elapsed time. [1 mark]

Question 22 | 2 marks

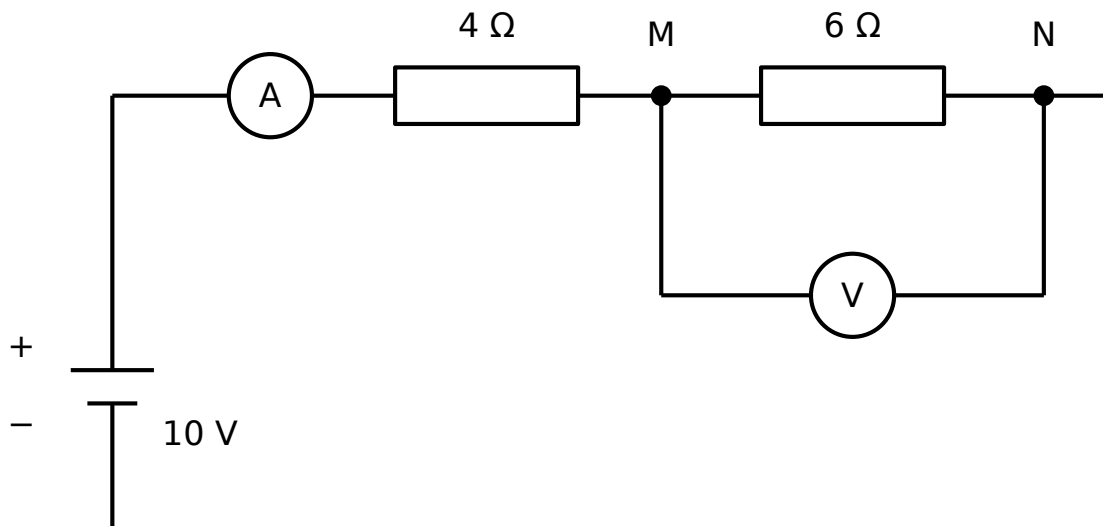
special-39-03-q22

Device U transfers 240 J while 40 C passes through it; device V transfers 180 J while 20 C passes. (i) Calculate their potential differences. [1 mark] (ii) A student says U must have the higher voltage because it transfers more energy. Evaluate the claim. [1 mark]

Question 23 | 2 marks

special-39-03-q23

In the supplied experimental circuit, both meters are ideal and the 10 V source is ideal. (i) Predict the ammeter and voltmeter readings. [1 mark] (ii) A student divides the voltmeter reading by the ammeter reading and labels the result “the total resistance connected to the source”. Explain the error and give the resistance that this ratio actually measures. [1 mark]



Ideal meters; dots mark connected nodes.

Question 24 | 2 marks

special-39-03-q24

For a resistor carrying a steady current I for time t at voltage V , (i) use $W = VQ$ and $Q = It$ to obtain the heat transferred, assuming all electrical energy becomes heat. [1 mark] (ii) Apply Ohm's law to express this heat in terms of I , R and t , and state what must stay fixed when claiming that heat is proportional to resistance. [1 mark]

Question 25 | 2 marks

special-39-03-q25

Two settings of a device both meet the same processing requirement: setting X draws 900 W for 8 min and setting Y draws 600 W for 13 min. The tariff is ₹8 per kWh with no other charges. (i) Which uses less energy, and by how much per run? [1 mark] (ii) Find the saving for 50 runs using the better setting. [1 mark]

Question 26 | 2 marks

special-39-03-q26

A student compares two different wire materials at the same voltage. Wire A is twice as long as B and both have the same diameter. A carries less current, so the student concludes that A's material has higher resistivity. (i) Is that conclusion guaranteed? Explain. [1 mark] (ii) Specify a controlled replacement comparison that would let current alone rank the two resistivities. [1 mark]

Section C

Question 27 | 3 marks

special-39-03-q27

A uniform wire has resistivity $1.2 \times 10^{-6} \Omega \text{ m}$, length 0.80 m and area 0.20 mm^2 . It forms a constant-resistance element. (i) Find its resistance. [1 mark] (ii) A replacement of the same material and length has twice the diameter. Decide whether it uses less energy during the same time at the same voltage, supporting your answer with resistance and energy ratios. [2 marks]

Question 28 | 3 marks

special-39-03-q28

An ideal source maintains 12 V across a purely resistive element for 60 s. In a two-stage approximation to its measured behaviour, its resistance is 6Ω for the first 30 s and 12Ω for the next 30 s; treat each stage as constant and the transition as instantaneous. All electrical input energy becomes generated heat. This stated change overrides the paper's usual fixed-resistance assumption.

(i) Calculate the heat generated in each stage and the total. [1 mark]

(ii) A student correctly finds the time-average current to be 1.5 A and the time-average resistance to be 9Ω , then reports $H = (1.5)^2 \times 9 \times 60 = 1215 \text{ J}$. Diagnose the error and quantify the discrepancy from the correct total. [1 mark]

(iii) Another student proposes $H = 12 \times 1.5 \times 60$ using the same average current. Is this method valid for this run? Explain using total charge or the two stage contributions, not calculus. [1 mark]

Question 29 | 3 marks

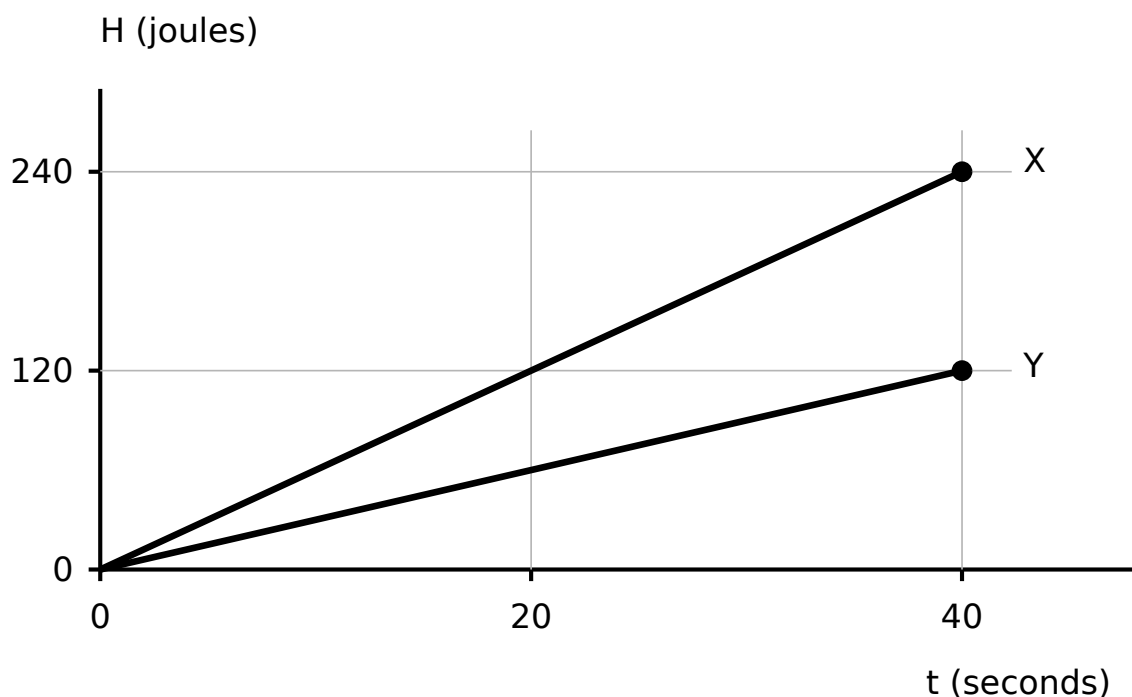
special-39-03-q29

A fixed-resistance mat is rated 90 W at 18 V. A technician wants it to generate 5400 J in no more than 90 s. Only ideal 12 V and 15 V sources are available, one used at a time. (i) Infer the mat resistance. [1 mark] (ii) Determine which source, if either, meets the target, calculating the required time for each. [2 marks]

Question 30 | 3 marks

special-39-03-q30

The graph shows cumulative heat generated by resistors X and Y during a controlled test; the two straight lines have been measured independently. X is 6Ω and Y is 12Ω . One student says the graph proves equal currents; another says it is consistent with equal voltages. (i) Read the heat ratio $H_X:H_Y$ at 40 s. [1 mark] (ii) Use the heat and time to test both statements and identify the consistent condition, including its common electrical value. [2 marks]



Question 31 | 3 marks

special-39-03-q31

A $3\ \Omega$ resistor is in series with two $6\ \Omega$ resistors in parallel, across an ideal $12\ \text{V}$ source. All resistances are fixed. One of the $6\ \Omega$ branches becomes open-circuit. (i) Find the current through the remaining $6\ \Omega$ resistor before and after this fault. [2 marks] (ii) Decide whether the heat it generates in a given time increases or decreases, and give the factor. [1 mark]

Question 32 | 3 marks

special-39-03-q32

An ideal supply has two independent operating limits: instantaneous output power must not exceed $100\ \text{W}$, and total delivered energy during a session must not exceed $40\ \text{Wh}$. A $60\ \text{W}$ task runs for $30\ \text{min}$ and a $50\ \text{W}$ task runs for $10\ \text{min}$; each draws its stated constant power while on. (i) Can they run simultaneously for the first $10\ \text{min}$? [1 mark] (ii) Can both be completed one after the other within the energy limit? [1 mark] (iii) Explain why satisfying one of these limits does not automatically satisfy the other. [1 mark]

Question 33 | 3 marks

special-39-03-q33

A resistor was previously measured as $20\ \Omega$ using trusted instruments. In a new test at the same temperature, an accurate voltmeter is connected directly across it. Suspect ammeter A displays $0.125\ \text{A}$ at $2.0\ \text{V}$ and $0.250\ \text{A}$ at $4.0\ \text{V}$. Consider two explanations: the resistor has genuinely changed to a different fixed resistance and A is accurate; or the resistor is unchanged and A displays a constant multiple of the true current. Both meters have negligible loading, and readings are exact for this question.

- What resistance would be inferred if A's readings were assumed accurate? [1 mark]
- A is replaced, in the same series position, by independently calibrated ammeter B. With the resistor, temperature and measured voltages unchanged, B reads $0.100\ \text{A}$ at $2.0\ \text{V}$ and $0.200\ \text{A}$

at 4.0 V. Which explanation does this evidence support? Determine the factor by which A multiplies the true current. [1 mark]

(iii) A student argues that A must be accurate because it reads zero when no current flows and its displayed-current-versus-voltage plot is a straight line through the origin. Explain why neither observation rules out the error identified in (ii). [1 mark]

Section D

Question 34 | 5 marks

special-39-03-q34

A workshop compares two complete programmes that tests have shown give the same accepted drying result. Programme A draws 1.2 kW for 6 min, then 80 W for 15 min. Programme B draws 750 W for 10 min, then 30 W for 20 min. Powers are actual constant input powers; all unlisted intervals draw zero power. There are 24 runs per day. The daily energy budget is 3.30 kWh and the tariff is ₹9 per kWh, with no other charges.

- (i) Calculate the electrical energy of each complete programme per run. [2 marks]
- (ii) Select the programme that fits the daily energy budget and show the daily margin or excess for each. [2 marks]
- (iii) Find the daily cost saving from choosing the lower-energy programme, and explain why choosing solely by the largest stated power could mislead. [1 mark]

Question 35 | 5 marks

special-39-03-q35

A purely resistive portable warmer is rated 48 W at 12 V. Its resistance is assumed unchanged at other voltages. A proposed source maintains 9 V at the warmer. For a particular task, exactly three-quarters of the electrical input energy reaches the useful load; the rest is lost to the surroundings. The load needs 4050 J of useful heat and the deadline is 4 min.

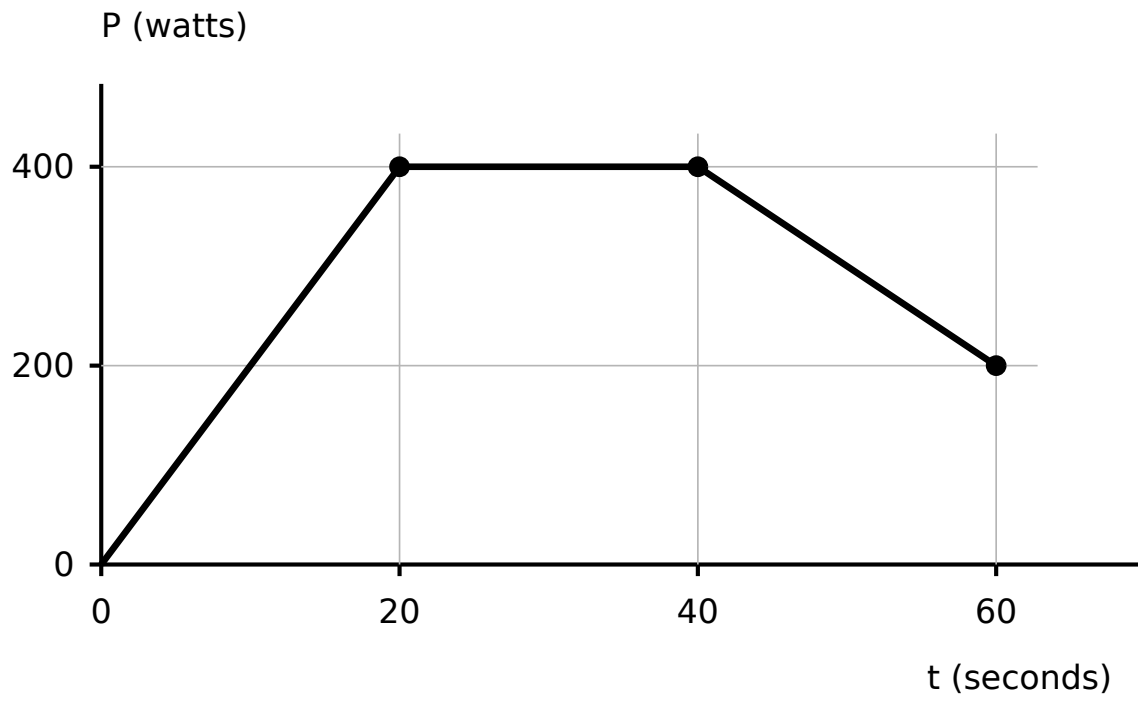
- (i) Infer the warmer's resistance and actual power at 9 V. [2 marks]
- (ii) Calculate the electrical energy required and the time at 9 V. Does the proposal meet the deadline? [2 marks]
- (iii) A student instead inserts 48 W into the time calculation because "the printed power cannot change". Explain quantitatively why that prediction is wrong. [1 mark]

Question 36 | 5 marks

special-39-03-q36

The figure shows the complete input-power schedule for a controlled resistive device during 0–60 s. The sloping segments are straight. The source can deliver at most 15 kJ during this run and at most 450 W at any instant. Use rectangles and triangles only; the energy of a straight ramp is its average endpoint power multiplied by its duration.

- (i) Calculate the energy used in each of the three plotted intervals and the total. [2 marks]
- (ii) Decide whether the plotted run satisfies both source limits, with evidence. [1 mark]
- (iii) If the original plotted schedule is followed without modification, at what time does the accumulated energy first reach 9 kJ? Show how you locate the time. [2 marks]



Section E

Question 37 | 4 marks

special-39-03-q37

A field recorder has a source that maintains 6 V and can deliver 1800 J of electrical energy before stopping. Each complete recording cycle draws 0.30 A for 20 s, then 0.05 A for 100 s. There is no off-cycle consumption and no loss between source and recorder; each stated current is constant. A logbook says that “the battery has 300 coulombs of electrons which the recorder will destroy”.

- (i) Find the charge passing through the recorder in one cycle. [1 mark]
- (ii) Determine the maximum number of complete cycles the energy supply permits, including the energy left after those complete cycles. [2 marks]
- (iii) Correct the logbook’s physical interpretation while explaining the origin of the number 300 C. [1 mark]

Question 38 | 4 marks

special-39-03-q38

A laboratory compares fixed resistors $A = 4\ \Omega$ and $B = 16\ \Omega$. During each 30 s test all electrical energy is measured as generated heat. In session 1 the source is adjusted separately so that each resistor carries 0.50 A. In session 2 each resistor is connected separately to the same ideal 8 V source. A report merges the sessions and concludes: “Greater resistance always generates more heat.” No temperature-rise comparisons are requested.

- (i) Find $H_A:H_B$ in session 1 and the heat in B. [1 mark]
- (ii) A student’s session-2 log records $H_A = 120\ \text{J}$ and $H_B = 480\ \text{J}$. Diagnose this log using the predicted heat ratio, and give the corrected heat in A. [1 mark]
- (iii) Rewrite the report’s conclusion so that it is valid for both sessions, explaining the different control variables. [1 mark]
- (iv) A third session doubles both A’s current and its test duration relative to session 1. The measured heat is eight times its session-1 value. A student concludes that doubling current alone causes eight times the heat. Evaluate this conclusion and specify a repeat that isolates the current effect, including the expected heat factor. [1 mark]

Question 39 | 4 marks

special-39-03-q39

A team needs a fixed-resistance element to generate exactly 1800 J in 60 s when connected directly to an ideal 12 V source. All input energy becomes generated heat. Three uniform candidate wires of the same stable alloy and temperature are available: A has length 0.50 m and area $0.10\ \text{mm}^2$; B has length 1.00 m and area $0.20\ \text{mm}^2$; C has length 0.50 m and area $0.20\ \text{mm}^2$. The alloy resistivity is $9.6 \times 10^{-7}\ \Omega\ \text{m}$. The wires are alternatives, not connected together.

- (i) Determine the required resistance from the task. [1 mark]
- (ii) Calculate the three candidate resistances and choose all candidates that meet the task exactly. [2 marks]
- (iii) A student says B must be rejected because it is longer than A. Explain why that reasoning is incomplete. [1 mark]

Detailed solutions and marking guidance

Original model answers, numbered to match the question paper. These are not an official marking scheme.

Solution 1 — special-39-03-q01

Correct option: B

Correct option: B. The magnitude of net charge is $Q = Ne = (5.0 \times 10^{18})(1.6 \times 10^{-19}) = 0.80 \text{ C}$. Hence $I = Q/t = 0.80/4.0 = 0.20 \text{ A}$. The electron's negative sign concerns direction, not the requested magnitude. The 0.80 value is charge, 3.2 multiplies rather than divides by time, and 1.25 is not Q/t . [1 mark]

Mark allocation

The magnitude of net charge is $Q = Ne = (5.0 \times 10^{18})(1.6 \times 10^{-19}) = 0.80 \text{ C}$. Hence $I = Q/t = 0.80/4.0 = 0.20 \text{ A}$. The electron's negative sign concerns direction, not the requested magnitude. The 0.80 value is charge, 3.2 multiplies rather than divides by time, and 1.25 is not Q/t . [1 marks]

Solution 2 — special-39-03-q02

Correct option: D

Correct option: D. One volt means one joule of work per coulomb transferred between the two points: $1 \text{ V} = 1 \text{ J/C}$. Option A defines an ampere; B defines a watt; C divides work by current rather than by charge and does not define a volt. [1 mark]

Mark allocation

One volt means one joule of work per coulomb transferred between the two points: $1 \text{ V} = 1 \text{ J/C}$. Option A defines an ampere; B defines a watt; C divides work by current rather than by charge and does not define a volt. [1 marks]

Solution 3 — special-39-03-q03

Correct option: A

Correct option: A. $I_X = 90/30 = 3 \text{ A}$ and $I_Y = 120/60 = 2 \text{ A}$. Hence X has 1.5 times the current of Y despite transferring less total charge. [1 mark]

Mark allocation

$I_X = 90/30 = 3 \text{ A}$ and $I_Y = 120/60 = 2 \text{ A}$. Hence X has 1.5 times the current of Y despite transferring less total charge. [1 marks]

Solution 4 — special-39-03-q04

Correct option: C

Correct option: C. At 6 V the graph gives $I_X = 0.60 \text{ A}$ and $I_Y = 0.30 \text{ A}$. $E_X = 6 \times 0.60 \times 600 = 2160 \text{ J}$; $E_Y = 6 \times 0.30 \times 600 = 1080 \text{ J}$. Only Y stays within 1500 J. [1 mark]

Mark allocation

At 6 V the graph gives $I_X = 0.60 \text{ A}$ and $I_Y = 0.30 \text{ A}$. $E_X = 6 \times 0.60 \times 600 = 2160 \text{ J}$; $E_Y = 6 \times 0.30 \times 600 = 1080 \text{ J}$. Only Y stays within 1500 J. [1 marks]

Solution 5 — special-39-03-q05

Correct option: B

Correct option: B. Y has twice the length but four times the area, so $R_Y/R_X = 2/4 = 1/2$. At equal I and t , $H = I^2Rt$ gives $H_Y/H_X = 1/2$. The material and temperature are unchanged, so resistivity is unchanged. [1 mark]

Mark allocation

Y has twice the length but four times the area, so $R_Y/R_X = 2/4 = 1/2$. At equal I and t , $H = I^2Rt$ gives $H_Y/H_X = 1/2$. The material and temperature are unchanged, so resistivity is unchanged. [1 marks]

Solution 6 — special-39-03-q06

Correct option: D

Correct option: D. $EA = 800 \times 9/60 = 120$ Wh. $EB = 500 \times 12/60 + 20 \times 15/60 = 105$ Wh. Trial B uses 15 Wh less; the same accepted outcome permits a meaningful energy comparison. [1 mark]

Mark allocation

$EA = 800 \times 9/60 = 120$ Wh. $EB = 500 \times 12/60 + 20 \times 15/60 = 105$ Wh. Trial B uses 15 Wh less; the same accepted outcome permits a meaningful energy comparison. [1 marks]

Solution 7 — special-39-03-q07

Correct option: A

Correct option: A. $R = 12^2/60 = 2.4 \Omega$. At 6 V, $P = 6^2/2.4 = 15$ W. The time is $900/15 = 60$ s, not 15 s from the rated power. [1 mark]

Mark allocation

$R = 12^2/60 = 2.4 \Omega$. At 6 V, $P = 6^2/2.4 = 15$ W. The time is $900/15 = 60$ s, not 15 s from the rated power. [1 marks]

Solution 8 — special-39-03-q08

Correct option: C

Correct option: C. The first, second and fourth pairs all give $V/I = 10 \Omega$. At 4.5 V the consistent current is $4.5/10 = 0.45$ A. Replacing 0.90 A by 0.45 A restores the constant ratio. [1 mark]

Mark allocation

The first, second and fourth pairs all give $V/I = 10 \Omega$. At 4.5 V the consistent current is $4.5/10 = 0.45$ A. Replacing 0.90 A by 0.45 A restores the constant ratio. [1 marks]

Solution 9 — special-39-03-q09

Correct option: B

Correct option: B. B is correct. Resistivity is a property of a material at a specified temperature, and can change when that temperature changes. The room-temperature value is not guaranteed to describe the hot element. No universal zero value follows from heating, and dividing by length does not supply temperature dependence. The question supplies no law for

calculating the hot value or its direction of change. [1 mark]

Mark allocation

B is correct. Resistivity is a property of a material at a specified temperature, and can change when that temperature changes. The room-temperature value is not guaranteed to describe the hot element. No universal zero value follows from heating, and dividing by length does not supply temperature dependence. The question supplies no law for calculating the hot value or its direction of change. [1 marks]

Solution 10 — special-39-03-q10

Correct option: D

Correct option: D. With S open only the $12\ \Omega$ branch conducts, drawing $12/12 = 1\text{ A}$. Closing S adds a $6\ \Omega$ branch drawing 2 A ; total current becomes 3 A , but the $12\ \Omega$ branch remains across 12 V and stays at 1 A . [1 mark]

Mark allocation

With S open only the $12\ \Omega$ branch conducts, drawing $12/12 = 1\text{ A}$. Closing S adds a $6\ \Omega$ branch drawing 2 A ; total current becomes 3 A , but the $12\ \Omega$ branch remains across 12 V and stays at 1 A . [1 marks]

Solution 11 — special-39-03-q11

Correct option: A

Correct option: A. $H_{\text{second}}/H_{\text{first}} = (1/2)^2 \times 2 = 1/2$. Doubling the duration does not compensate for reducing current by a factor of two. [1 mark]

Mark allocation

$H_{\text{second}}/H_{\text{first}} = (1/2)^2 \times 2 = 1/2$. Doubling the duration does not compensate for reducing current by a factor of two. [1 marks]

Solution 12 — special-39-03-q12

Correct option: C

Correct option: C. For individual resistance R , $R_s = 2R$ and $R_p = R/2$. At the same voltage $P_{\text{parallel}}/P_{\text{series}} = R_s/R_p = 4$. Equal energies require $t_{\text{parallel}} = t/4$. [1 mark]

Mark allocation

For individual resistance R , $R_s = 2R$ and $R_p = R/2$. At the same voltage $P_{\text{parallel}}/P_{\text{series}} = R_s/R_p = 4$. Equal energies require $t_{\text{parallel}} = t/4$. [1 marks]

Solution 13 — special-39-03-q13

Correct option: B

Correct option: B. In steady operation 2 C enters and 2 C leaves each second, so charge is not consumed. The transferred energy per coulomb is $10/2 = 5\text{ J/C}$, consistent with a 5 V drop across the resistor. [1 mark]

Mark allocation

In steady operation 2 C enters and 2 C leaves each second, so charge is not consumed. The transferred energy per coulomb is $10/2 = 5 \text{ J/C}$, consistent with a 5 V drop across the resistor. [1 marks]

Solution 14 — special-39-03-q14

Correct option: D

Correct option: D. The meter overreads by 0.4 V. True voltage = $2.4 - 0.4 = 2.0 \text{ V}$ and $R = 2.0/0.50 = 4.0 \Omega$. Adding the offset or using the uncorrected reading would overestimate R. [1 mark]

Mark allocation

The meter overreads by 0.4 V. True voltage = $2.4 - 0.4 = 2.0 \text{ V}$ and $R = 2.0/0.50 = 4.0 \Omega$. Adding the offset or using the uncorrected reading would overestimate R. [1 marks]

Solution 15 — special-39-03-q15

Correct option: A

Correct option: A. Suitable heating alloys generally have relatively high resistivity and do not oxidise readily at high temperatures. Low resistance alone or rapid oxidation would not justify their use. [1 mark]

Mark allocation

Suitable heating alloys generally have relatively high resistivity and do not oxidise readily at high temperatures. Low resistance alone or rapid oxidation would not justify their use. [1 marks]

Solution 16 — special-39-03-q16

Correct option: C

Correct option: C. From the graph $E = 600 \times 2/60 + 200 \times 4/60 + 0 \times 2/60 = 33\frac{1}{3} \text{ Wh}$. The alternative uses $300 \times 8/60 = 40 \text{ Wh}$. The plotted schedule uses $6\frac{2}{3} \text{ Wh}$ less. [1 mark]

Mark allocation

From the graph $E = 600 \times 2/60 + 200 \times 4/60 + 0 \times 2/60 = 33\frac{1}{3} \text{ Wh}$. The alternative uses $300 \times 8/60 = 40 \text{ Wh}$. The plotted schedule uses $6\frac{2}{3} \text{ Wh}$ less. [1 marks]

Solution 17 — special-39-03-q17

Correct option: D

Correct option: D. A is false: doubling R at fixed V and t halves heat. R is true, since $H = VIt$ and $I = V/R$ give $H = V^2t/R$. Thus option D applies. [1 mark]

Mark allocation

A is false: doubling R at fixed V and t halves heat. R is true, since $H = VIt$ and $I = V/R$ give $H = V^2t/R$. Thus option D applies. [1 marks]

Solution 18 — special-39-03-q18

Correct option: B

Correct option: B. A is true because kWh is power multiplied by time. R is also true, but how an ammeter is connected does not explain the dimension of kWh. Hence B. [1 mark]

Mark allocation

A is true because kWh is power multiplied by time. R is also true, but how an ammeter is connected does not explain the dimension of kWh. Hence B. [1 marks]

Solution 19 — special-39-03-q19

Correct option: A

Correct option: A. Both statements are true. No steady accumulation means equal entering and leaving charge per unit time; this directly explains the equal series currents. Energy is transferred but charge is conserved. Hence A. [1 mark]

Mark allocation

Both statements are true. No steady accumulation means equal entering and leaving charge per unit time; this directly explains the equal series currents. Energy is transferred but charge is conserved. Hence A. [1 marks]

Solution 20 — special-39-03-q20

Correct option: C

Correct option: C. A is true: $E_{\text{new}}/E_{\text{reference}} = (1/2)^2 \times 2 = 1/2$. R is false because the heater is explicitly fixed-resistance; lowering voltage lowers current, not its resistance. Hence C. [1 mark]

Mark allocation

A is true: $E_{\text{new}}/E_{\text{reference}} = (1/2)^2 \times 2 = 1/2$. R is false because the heater is explicitly fixed-resistance; lowering voltage lowers current, not its resistance. Hence C. [1 marks]

Solution 21 — special-39-03-q21

(i) $Q = 0.40 \times 30 + 0.10 \times 90 = 12 + 9 = 21$ C. [1 mark]

(ii) Total time = 120 s. Average current = total charge/total time = $21/120 = 0.175$ A. The proposed 0.25 A is wrong because the lower-current stage lasts three times as long. [1 mark]

Mark allocation

(i) $Q = 0.40 \times 30 + 0.10 \times 90 = 12 + 9 = 21$ C. [1 marks]

(ii) Total time = 120 s. Average current = total charge/total time = $21/120 = 0.175$ A. The proposed 0.25 A is wrong because the lower-current stage lasts three times as long. [1 marks]

Solution 22 — special-39-03-q22

(i) $V_U = 240/40 = 6$ V; $V_V = 180/20 = 9$ V. Award 0.5 mark for each. [1 mark]

(ii) The claim is false: V transfers 9 J/C, compared with U's 6 J/C. More total energy does not imply a higher voltage when different amounts of charge pass. [1 mark]

Mark allocation

- (i) $V_U = 240/40 = 6 \text{ V}$; $V_V = 180/20 = 9 \text{ V}$. Award 0.5 mark for each. [1 marks]
- (ii) The claim is false: V transfers 9 J/C, compared with U's 6 J/C. More total energy does not imply a higher voltage when different amounts of charge pass. [1 marks]

Solution 23 — special-39-03-q23

- (i) The resistors are in series: $R_{\text{total}} = 4 + 6 = 10 \Omega$, so A reads $10/10 = 1 \text{ A}$. The voltmeter spans only the 6Ω resistor and reads $1 \times 6 = 6 \text{ V}$. Award 0.5 each. [1 mark]
- (ii) The ratio is $6 \text{ V}/1 \text{ A} = 6 \Omega$, the resistance between the voltmeter's connection nodes, not total resistance. Total resistance is 10Ω ; its calculation requires the source's full 10 V. [1 mark]

Mark allocation

- (i) The resistors are in series: $R_{\text{total}} = 4 + 6 = 10 \Omega$, so A reads $10/10 = 1 \text{ A}$. The voltmeter spans only the 6Ω resistor and reads $1 \times 6 = 6 \text{ V}$. Award 0.5 each. [1 marks]
- (ii) The ratio is $6 \text{ V}/1 \text{ A} = 6 \Omega$, the resistance between the voltmeter's connection nodes, not total resistance. Total resistance is 10Ω ; its calculation requires the source's full 10 V. [1 marks]

Solution 24 — special-39-03-q24

- (i) $H = W = VQ = V(It) = VIt$, for the purely resistive device. [1 mark]
- (ii) $V = IR$ gives $H = (IR)It = I^2Rt$. Heat is proportional to R only if current I and duration t are fixed. Award 0.5 for the expression and 0.5 for both conditions. [1 mark]

Mark allocation

- (i) $H = W = VQ = V(It) = VIt$, for the purely resistive device. [1 marks]
- (ii) $V = IR$ gives $H = (IR)It = I^2Rt$. Heat is proportional to R only if current I and duration t are fixed. Award 0.5 for the expression and 0.5 for both conditions. [1 marks]

Solution 25 — special-39-03-q25

- (i) $EX = 0.9 \times 8/60 = 0.12 \text{ kWh}$; $EY = 0.6 \times 13/60 = 0.13 \text{ kWh}$. X uses 0.01 kWh less per run despite its higher power. [1 mark]
- (ii) Energy saving over 50 runs = $50 \times 0.01 = 0.50 \text{ kWh}$. Money saved = $0.50 \times ₹8 = ₹4$. [1 mark]

Mark allocation

- (i) $EX = 0.9 \times 8/60 = 0.12 \text{ kWh}$; $EY = 0.6 \times 13/60 = 0.13 \text{ kWh}$. X uses 0.01 kWh less per run despite its higher power. [1 marks]
- (ii) Energy saving over 50 runs = $50 \times 0.01 = 0.50 \text{ kWh}$. Money saved = $0.50 \times ₹8 = ₹4$. [1 marks]

Solution 26 — special-39-03-q26

- (i) No. Lower current implies higher resistance at the same voltage, but $R = \rho L/A$. A's greater length can itself cause its higher resistance, even for equal or somewhat lower resistivity. [1 mark]
- (ii) Compare wires of the two materials with equal lengths and equal cross-sectional areas, at the same temperature and voltage. Then R is proportional to ρ , so the lower-current wire has

higher resistivity. [1 mark]

Mark allocation

(i) No. Lower current implies higher resistance at the same voltage, but $R = \rho L/A$. A's greater length can itself cause its higher resistance, even for equal or somewhat lower resistivity. [1 marks]

(ii) Compare wires of the two materials with equal lengths and equal cross-sectional areas, at the same temperature and voltage. Then R is proportional to ρ , so the lower-current wire has higher resistivity. [1 marks]

Solution 27 — special-39-03-q27

(i) $0.20 \text{ mm}^2 = 2.0 \times 10^{-7} \text{ m}^2$. $R = \rho L/A = (1.2 \times 10^{-6} \times 0.80)/(2.0 \times 10^{-7}) = 4.8 \Omega$. [1 mark]

(ii) Doubling diameter multiplies area by four; replacement resistance is $R/4 = 1.2 \Omega$. Award 1 mark for this chain. [1 mark]

At fixed V and t , $E = V^2 t/R$. The replacement uses four times the original energy, not less. Its lower resistance draws more current. [1 mark]

Mark allocation

(i) $0.20 \text{ mm}^2 = 2.0 \times 10^{-7} \text{ m}^2$. $R = \rho L/A = (1.2 \times 10^{-6} \times 0.80)/(2.0 \times 10^{-7}) = 4.8 \Omega$. [1 marks]

(ii) Doubling diameter multiplies area by four; replacement resistance is $R/4 = 1.2 \Omega$. Award 1 mark for this chain. [1 marks]

At fixed V and t , $E = V^2 t/R$. The replacement uses four times the original energy, not less. Its lower resistance draws more current. [1 marks]

Solution 28 — special-39-03-q28

(i) First stage: $I_1 = 12/6 = 2 \text{ A}$, $H_1 = 12^2 \times 30/6 = 720 \text{ J}$. Second stage: $I_2 = 12/12 = 1 \text{ A}$, $H_2 = 12^2 \times 30/12 = 360 \text{ J}$. Total $H = 720 + 360 = 1080 \text{ J}$. Award 0.5 for both stage heats and 0.5 for the total with working. [1 mark]

(ii) The actual pairs are (2 A, 6 Ω) and (1 A, 12 Ω). $H = I^2 R t$ must use the matching current and resistance for each constant stage; squaring the average current and multiplying by the separately averaged resistance does not preserve these stagewise products. The 1215 J report exceeds the correct total by $1215 - 1080 = 135 \text{ J}$. Award 0.5 for identifying this invalid replacement and 0.5 for the excess. [1 mark]

(iii) Yes, because the voltage remains 12 V throughout. Total charge $Q = 2 \times 30 + 1 \times 30 = 90 \text{ C} = I_{\text{average}} \times 60$. Thus $H = VQ = 12 \times 90 = 1080 \text{ J} = 12 \times 1.5 \times 60$. Award 0.5 for the fixed-voltage/total-charge justification and 0.5 for the matching total. An average current is not inherently invalid; its use depends on the energy expression and controlled quantities. [1 mark]

Mark allocation

(i) First stage: $I_1 = 12/6 = 2 \text{ A}$, $H_1 = 12^2 \times 30/6 = 720 \text{ J}$. Second stage: $I_2 = 12/12 = 1 \text{ A}$, $H_2 = 12^2 \times 30/12 = 360 \text{ J}$. Total $H = 720 + 360 = 1080 \text{ J}$. Award 0.5 for both stage heats and 0.5 for the total with working. [1 marks]

(ii) The actual pairs are (2 A, 6 Ω) and (1 A, 12 Ω). $H = I^2 R t$ must use the matching current and resistance for each constant stage; squaring the average current and multiplying by the separately averaged resistance does not preserve these stagewise products. The 1215 J report exceeds the correct total by $1215 - 1080 = 135 \text{ J}$. Award 0.5 for identifying this invalid

replacement and 0.5 for the excess. [1 marks]

(iii) Yes, because the voltage remains 12 V throughout. Total charge $Q = 2 \times 30 + 1 \times 30 = 90$ C = laverage $\times 60$. Thus $H = VQ = 12 \times 90 = 1080$ J = $12 \times 1.5 \times 60$. Award 0.5 for the fixed-voltage/total-charge justification and 0.5 for the matching total. An average current is not inherently invalid; its use depends on the energy expression and controlled quantities. [1 marks]

Solution 29 — special-39-03-q29

(i) $R = 18^2/90 = 3.6 \Omega$. [1 mark]

(ii) At 12 V, $P = 12^2/3.6 = 40$ W and time = $5400/40 = 135$ s, which misses the 90 s deadline. [1 mark]

At 15 V, $P = 15^2/3.6 = 62.5$ W and time = $5400/62.5 = 86.4$ s. Only the 15 V source meets the deadline. [1 mark]

Mark allocation

(i) $R = 18^2/90 = 3.6 \Omega$. [1 marks]

(ii) At 12 V, $P = 12^2/3.6 = 40$ W and time = $5400/40 = 135$ s, which misses the 90 s deadline. [1 marks]

At 15 V, $P = 15^2/3.6 = 62.5$ W and time = $5400/62.5 = 86.4$ s. Only the 15 V source meets the deadline. [1 marks]

Solution 30 — special-39-03-q30

(i) At 40 s, $H_X = 240$ J and $H_Y = 120$ J, so $H_X:H_Y = 2:1$. [1 mark]

(ii) $P_X = 240/40 = 6$ W and $P_Y = 120/40 = 3$ W. $I_X = \sqrt{(6/6)} = 1$ A and $I_Y = \sqrt{(3/12)} = 0.5$ A; they are not equal. An equal-current test would instead make the 12 Ω resistor produce twice as much heat. [1 mark]

$V_X = I_X R_X = 1 \times 6 = 6$ V and $V_Y = I_Y R_Y = 0.5 \times 12 = 6$ V. Thus the graph is consistent with equal 6 V supplies, not equal currents. [1 mark]

Mark allocation

(i) At 40 s, $H_X = 240$ J and $H_Y = 120$ J, so $H_X:H_Y = 2:1$. [1 marks]

(ii) $P_X = 240/40 = 6$ W and $P_Y = 120/40 = 3$ W. $I_X = \sqrt{(6/6)} = 1$ A and $I_Y = \sqrt{(3/12)} = 0.5$ A; they are not equal. An equal-current test would instead make the 12 Ω resistor produce twice as much heat. [1 marks]

$V_X = I_X R_X = 1 \times 6 = 6$ V and $V_Y = I_Y R_Y = 0.5 \times 12 = 6$ V. Thus the graph is consistent with equal 6 V supplies, not equal currents. [1 marks]

Solution 31 — special-39-03-q31

(i) Before the fault, parallel resistance is 3 Ω and total resistance 6 Ω . Source current is $12/6 = 2$ A, shared equally: each 6 Ω resistor carries 1 A. [1 mark]

After the fault, the path is 3 $\Omega + 6 \Omega = 9 \Omega$. The current in the remaining resistor is $12/9 = 4/3$ A. [1 mark]

(ii) At the same R and t, $H_{\text{after}}/H_{\text{before}} = (4/3)^2/1^2 = 16/9$. Its heat increases by a factor of 16/9 even though total source current falls. [1 mark]

Mark allocation

(i) Before the fault, parallel resistance is $3\ \Omega$ and total resistance $6\ \Omega$. Source current is $12/6 = 2$ A, shared equally: each $6\ \Omega$ resistor carries 1 A. [1 marks]

After the fault, the path is $3\ \Omega + 6\ \Omega = 9\ \Omega$. The current in the remaining resistor is $12/9 = 4/3$ A. [1 marks]

(ii) At the same R and t, $H_{\text{after}}/H_{\text{before}} = (4/3)^2/1^2 = 16/9$. Its heat increases by a factor of 16/9 even though total source current falls. [1 marks]

Solution 32 — special-39-03-q32

(i) No. Their simultaneous power is $60 + 50 = 110$ W, above 100 W. [1 mark]

(ii) Yes. Sequential power is at most 60 W and total energy is $60 \times 30/60 + 50 \times 10/60 = 30 + 8\frac{1}{3} = 38\frac{1}{3}$ Wh, below 40 Wh. [1 mark]

(iii) Power limits the rate of energy delivery; energy also depends on how long the tasks run. Short overlap can violate the power limit without exceeding the energy budget; a long low-power run can exceed an energy budget while staying below the power limit. [1 mark]

Mark allocation

(i) No. Their simultaneous power is $60 + 50 = 110$ W, above 100 W. [1 marks]

(ii) Yes. Sequential power is at most 60 W and total energy is $60 \times 30/60 + 50 \times 10/60 = 30 + 8\frac{1}{3} = 38\frac{1}{3}$ Wh, below 40 Wh. [1 marks]

(iii) Power limits the rate of energy delivery; energy also depends on how long the tasks run. Short overlap can violate the power limit without exceeding the energy budget; a long low-power run can exceed an energy budget while staying below the power limit. [1 marks]

Solution 33 — special-39-03-q33

(i) From A, $R = 2.0/0.125 = 16\ \Omega$; the second pair gives $4.0/0.250 = 16\ \Omega$ as well. The inferred value would therefore be $16\ \Omega$, not the earlier $20\ \Omega$. [1 mark]

(ii) Trusted B gives $2.0/0.100 = 20\ \Omega$ and $4.0/0.200 = 20\ \Omega$, supporting unchanged resistance and an error in A. $A/B = 0.125/0.100 = 0.250/0.200 = 1.25$, so A displays 1.25 times the true current, an overreading of 25%. Award 0.5 for the diagnosis from B's resistance and 0.5 for the consistent factor. [1 mark]

(iii) Multiplying a zero true current by 1.25 still gives zero, so a correct zero reading does not establish a correct scale. Also, for the $20\ \Omega$ resistor, $I_{\text{true}} = V/20$ and $I_A = 1.25V/20 = V/16$: multiplying every current by the same factor preserves a straight line through the origin but changes its slope. Award 0.5 for each explanation. The independently calibrated comparison, not either shape/zero test alone, exposes this error. [1 mark]

Mark allocation

(i) From A, $R = 2.0/0.125 = 16\ \Omega$; the second pair gives $4.0/0.250 = 16\ \Omega$ as well. The inferred value would therefore be $16\ \Omega$, not the earlier $20\ \Omega$. [1 marks]

(ii) Trusted B gives $2.0/0.100 = 20\ \Omega$ and $4.0/0.200 = 20\ \Omega$, supporting unchanged resistance and an error in A. $A/B = 0.125/0.100 = 0.250/0.200 = 1.25$, so A displays 1.25 times the true current, an overreading of 25%. Award 0.5 for the diagnosis from B's resistance and 0.5 for the consistent factor. [1 marks]

(iii) Multiplying a zero true current by 1.25 still gives zero, so a correct zero reading does not establish a correct scale. Also, for the $20\ \Omega$ resistor, $I_{\text{true}} = V/20$ and $I_A = 1.25V/20 = V/16$: multiplying every current by the same factor preserves a straight line through the origin but changes its slope. Award 0.5 for each explanation. The independently calibrated comparison, not either shape/zero test alone, exposes this error. [1 marks]

Solution 34 — special-39-03-q34

(i) A: $1.2 \times 6/60 + 0.080 \times 15/60 = 0.120 + 0.020 = 0.140$ kWh per run. [1 mark]

B: $0.750 \times 10/60 + 0.030 \times 20/60 = 0.125 + 0.010 = 0.135$ kWh per run. [1 mark]

(ii) A uses $24 \times 0.140 = 3.36$ kWh/day, exceeding the budget by $3.36 - 3.30 = 0.06$ kWh. [1 mark]

B uses $24 \times 0.135 = 3.24$ kWh/day, leaving $3.30 - 3.24 = 0.06$ kWh. Choose B, since only B fits the daily budget and both meet the process requirement. [1 mark]

(iii) Daily saving = $(3.36 - 3.24) \times ₹9 = ₹1.08$. Award 0.5 for the cost and 0.5 for explaining that energy depends on each power AND its duration, including after-runs; the largest power by itself is insufficient. [1 mark]

Mark allocation

(i) A: $1.2 \times 6/60 + 0.080 \times 15/60 = 0.120 + 0.020 = 0.140$ kWh per run. [1 marks]

B: $0.750 \times 10/60 + 0.030 \times 20/60 = 0.125 + 0.010 = 0.135$ kWh per run. [1 marks]

(ii) A uses $24 \times 0.140 = 3.36$ kWh/day, exceeding the budget by $3.36 - 3.30 = 0.06$ kWh. [1 marks]

B uses $24 \times 0.135 = 3.24$ kWh/day, leaving $3.30 - 3.24 = 0.06$ kWh. Choose B, since only B fits the daily budget and both meet the process requirement. [1 marks]

(iii) Daily saving = $(3.36 - 3.24) \times ₹9 = ₹1.08$. Award 0.5 for the cost and 0.5 for explaining that energy depends on each power AND its duration, including after-runs; the largest power by itself is insufficient. [1 marks]

Solution 35 — special-39-03-q35

(i) $R = 12^2/48 = 3\ \Omega$. [1 mark]

At 9 V, actual input power $P = 9^2/3 = 27$ W (current 3 A). The 48 W rating applies at 12 V, not every source voltage. [1 mark]

(ii) Since useful heat = $(3/4) \times$ input energy, required electrical input = $4050/(3/4) = 5400$ J. [1 mark]

Time = $5400/27 = 200$ s = 3 min 20 s. The deadline is 240 s, so the proposal meets it with 40 s to spare. [1 mark]

(iii) Using 48 W would predict $5400/48 = 112.5$ s. At 9 V that interval supplies only $27 \times 112.5 = 3037.5$ J electrically, or 2278.125 J usefully, below 4050 J. Power varies as V^2 for the stated fixed resistance; the printed rating is not constant-power regulation. [1 mark]

Mark allocation

(i) $R = 12^2/48 = 3\ \Omega$. [1 marks]

At 9 V, actual input power $P = 9^2/3 = 27$ W (current 3 A). The 48 W rating applies at 12 V, not every source voltage. [1 marks]

(ii) Since useful heat = $(3/4) \times$ input energy, required electrical input = $4050/(3/4) = 5400$ J. [1 marks]

Time = $5400/27 = 200$ s = 3 min 20 s. The deadline is 240 s, so the proposal meets it with 40 s to spare. [1 marks]

(iii) Using 48 W would predict $5400/48 = 112.5$ s. At 9 V that interval supplies only $27 \times 112.5 = 3037.5$ J electrically, or 2278.125 J usefully, below 4050 J. Power varies as V^2 for the stated fixed resistance; the printed rating is not constant-power regulation. [1 marks]

Solution 36 — special-39-03-q36

(i) From 0–20 s the triangular region gives $(1/2) \times 20 \times 400 = 4000$ J. From 20–40 s the rectangle gives $20 \times 400 = 8000$ J. Award 0.5 each. [1 mark]

From 40–60 s the trapezium gives $[(400 + 200)/2] \times 20 = 6000$ J. Total = $4000 + 8000 + 6000 = 18000$ J = 18 kJ. Award 0.5 each for the final interval and total. [1 mark]

(ii) Peak power is 400 W, below 450 W, so the power limit is satisfied. Total energy 18 kJ exceeds 15 kJ by 3 kJ, so the full run is not feasible within the energy limit. Award 0.5 for each test. [1 mark]

(iii) At 20 s, accumulated energy is 4 kJ; at 40 s it is 12 kJ. Thus 9 kJ is first reached during the constant 400 W interval, not on either ramp. [1 mark]

An additional $9000 - 4000 = 5000$ J takes $5000/400 = 12.5$ s after 20 s. The time from the start is therefore 32.5 s. [1 mark]

Mark allocation

(i) From 0–20 s the triangular region gives $(1/2) \times 20 \times 400 = 4000$ J. From 20–40 s the rectangle gives $20 \times 400 = 8000$ J. Award 0.5 each. [1 marks]

From 40–60 s the trapezium gives $[(400 + 200)/2] \times 20 = 6000$ J. Total = $4000 + 8000 + 6000 = 18000$ J = 18 kJ. Award 0.5 each for the final interval and total. [1 marks]

(ii) Peak power is 400 W, below 450 W, so the power limit is satisfied. Total energy 18 kJ exceeds 15 kJ by 3 kJ, so the full run is not feasible within the energy limit. Award 0.5 for each test. [1 marks]

(iii) At 20 s, accumulated energy is 4 kJ; at 40 s it is 12 kJ. Thus 9 kJ is first reached during the constant 400 W interval, not on either ramp. [1 marks]

An additional $9000 - 4000 = 5000$ J takes $5000/400 = 12.5$ s after 20 s. The time from the start is therefore 32.5 s. [1 marks]

Solution 37 — special-39-03-q37

(i) $Q_{\text{cycle}} = 0.30 \times 20 + 0.05 \times 100 = 6 + 5 = 11$ C. [1 mark]

(ii) $E_{\text{cycle}} = VQ = 6 \times 11 = 66$ J. Since $1800/66 = 27.27\dots$, only 27 complete cycles are possible; a fractional cycle does not count. [1 mark]

Energy used = $27 \times 66 = 1782$ J, leaving 18 J. A 28th full cycle needs 66 J, so the remainder is insufficient. [1 mark]

(iii) $1800 \text{ J}/6 \text{ J per C} = 300$ C is the total charge that can pass through the external circuit under this fixed-voltage energy model. Electrons are not destroyed; the source transfers stored energy as charge circulates. [1 mark]

Mark allocation

(i) $Q_{\text{cycle}} = 0.30 \times 20 + 0.05 \times 100 = 6 + 5 = 11 \text{ C}$. [1 marks]

(ii) $E_{\text{cycle}} = VQ = 6 \times 11 = 66 \text{ J}$. Since $1800/66 = 27.27\dots$, only 27 complete cycles are possible; a fractional cycle does not count. [1 marks]

Energy used = $27 \times 66 = 1782 \text{ J}$, leaving 18 J. A 28th full cycle needs 66 J, so the remainder is insufficient. [1 marks]

(iii) $1800 \text{ J}/6 \text{ J per C} = 300 \text{ C}$ is the total charge that can pass through the external circuit under this fixed-voltage energy model. Electrons are not destroyed; the source transfers stored energy as charge circulates. [1 marks]

Solution 38 — special-39-03-q38

(i) At equal I and t , $H_A:H_B = R_A:R_B = 4:16 = 1:4$. $H_B = 0.50^2 \times 16 \times 30 = 120 \text{ J}$. Award 0.5 each. [1 mark]

(ii) At equal V and t , $H_A:H_B = R_B:R_A = 16:4 = 4:1$, not the logged 1:4. $H_A = 8^2 \times 30/4 = 480 \text{ J}$ and $H_B = 8^2 \times 30/16 = 120 \text{ J}$: the entries are interchanged. Award 0.5 for diagnosing the reversed allocation using the ratio and 0.5 for the corrected 480 J with working. [1 mark]

(iii) At fixed current and time, larger resistance generates more heat; at fixed voltage and time, larger resistance generates less heat because the current decreases. The sessions cannot be merged into an unconditional rule about resistance. [1 mark]

(iv) The eightfold result is consistent with $H \propto I^2t$: doubling current gives factor 4 and doubling duration gives factor 2, together 8 (30 J becomes 240 J). It does not isolate current because duration also changed. Repeat with A at 1.0 A for the original 30 s, keeping resistance unchanged; expected heat is four times the original, 120 J. Award 0.5 for identifying the time confounder and 0.5 for the controlled repeat and factor 4. [1 mark]

Mark allocation

(i) At equal I and t , $H_A:H_B = R_A:R_B = 4:16 = 1:4$. $H_B = 0.50^2 \times 16 \times 30 = 120 \text{ J}$. Award 0.5 each. [1 marks]

(ii) At equal V and t , $H_A:H_B = R_B:R_A = 16:4 = 4:1$, not the logged 1:4. $H_A = 8^2 \times 30/4 = 480 \text{ J}$ and $H_B = 8^2 \times 30/16 = 120 \text{ J}$: the entries are interchanged. Award 0.5 for diagnosing the reversed allocation using the ratio and 0.5 for the corrected 480 J with working. [1 marks]

(iii) At fixed current and time, larger resistance generates more heat; at fixed voltage and time, larger resistance generates less heat because the current decreases. The sessions cannot be merged into an unconditional rule about resistance. [1 marks]

(iv) The eightfold result is consistent with $H \propto I^2t$: doubling current gives factor 4 and doubling duration gives factor 2, together 8 (30 J becomes 240 J). It does not isolate current because duration also changed. Repeat with A at 1.0 A for the original 30 s, keeping resistance unchanged; expected heat is four times the original, 120 J. Award 0.5 for identifying the time confounder and 0.5 for the controlled repeat and factor 4. [1 marks]

Solution 39 — special-39-03-q39

(i) Required power = $1800/60 = 30 \text{ W}$. Required resistance = $V^2/P = 12^2/30 = 4.8 \Omega$. [1 mark]

(ii) Using $0.10 \text{ mm}^2 = 1.0 \times 10^{-7} \text{ m}^2$ and $0.20 \text{ mm}^2 = 2.0 \times 10^{-7} \text{ m}^2$, $R_A = (9.6 \times 10^{-7} \times 0.50)/(1.0 \times 10^{-7}) = 4.8 \Omega$; $R_B = (9.6 \times 10^{-7} \times 1.00)/(2.0 \times 10^{-7}) = 4.8 \Omega$; $R_C = 2.4 \Omega$. Award 1 mark for all three. [1 mark]

A and B both meet the $4.8\ \Omega$ target and each generates 1800 J in 60 s. C generates $12^2 \times 60/2.4 = 3600$ J, twice the target, so it does not meet the exact specification. [1 mark]

(iii) B has twice A's length but also twice its cross-sectional area; the two effects cancel in $R = \rho L/A$. Length alone cannot rank their resistances or heating at fixed voltage. [1 mark]

Mark allocation

(i) Required power = $1800/60 = 30$ W. Required resistance = $V^2/P = 12^2/30 = 4.8\ \Omega$. [1 marks]

(ii) Using $0.10\ \text{mm}^2 = 1.0 \times 10^{-7}\ \text{m}^2$ and $0.20\ \text{mm}^2 = 2.0 \times 10^{-7}\ \text{m}^2$, $R_A = (9.6 \times 10^{-7} \times 0.50)/(1.0 \times 10^{-7}) = 4.8\ \Omega$; $R_B = (9.6 \times 10^{-7} \times 1.00)/(2.0 \times 10^{-7}) = 4.8\ \Omega$; $R_C = 2.4\ \Omega$. Award 1 mark for all three. [1 marks]

A and B both meet the $4.8\ \Omega$ target and each generates 1800 J in 60 s. C generates $12^2 \times 60/2.4 = 3600$ J, twice the target, so it does not meet the exact specification. [1 marks]

(iii) B has twice A's length but also twice its cross-sectional area; the two effects cancel in $R = \rho L/A$. Length alone cannot rank their resistances or heating at fixed voltage. [1 marks]

Source-pattern and rationale appendix

These questions are original. References identify supplied source patterns and curriculum connections, not official provenance for these questions or predictions of future examinations. No missing references or rationale have been inferred.

Question 1 — special-39-03-q01

Source patterns: ncert-11

NCERT references: §11.1, printed p.172: charge of an electron and $I = Q/t$

Competency classification: No

Competency marks: 0

Competency rationale: Foundational use of $Q = Ne$ followed by $I = Q/t$; the supplied electron charge removes any need for recalled numerical constants.

Rationale tags: foundational

Question 2 — special-39-03-q02

Source patterns: ncert-11

NCERT references: §11.2, printed p.173: definition of one volt, $1 \text{ V} = 1 \text{ J C}^{-1}$

Competency classification: No

Competency marks: 0

Competency rationale: Foundational recognition of the SI volt definition; no numerical W/Q substitution or competency credit.

Rationale tags: foundational

Question 3 — special-39-03-q03

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

Competency classification: No

Competency marks: 0

Competency rationale: Although presented as a comparison, this is only two direct $I = Q/t$ substitutions; no competency credit claimed.

Rationale tags: foundational

Question 4 — special-39-03-q04

Source patterns: pyq-2023-246eac3017f0793c27bba6a6
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Student must read the I -vertical graph, connect the current to energy, and test two alternatives against a cap.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 5 — special-39-03-q05

Source patterns: cbse_sqp_2018-19_electricity_q13
SQP_2023-24_q32
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–179
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 1

Competency rationale: Requires using diameter squared in area, then choosing the fixed-current heat relationship rather than the fixed-voltage one.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 6 — special-39-03-q06

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Must include a low-power after-run and compare complete schedules, not just nameplate power.

Rationale tags: evidence-and-reasoning

Question 7 — special-39-03-q07

Source patterns: pyq-2022-4f106d54aeecf637b6b663fc
ncert-11

NCERT references: §11.8, printed pp.191–192
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 1

Competency rationale: Transfers a rating to a new voltage and then uses the heat target; power is not held at its rating.

Rationale tags: evidence-and-reasoning

Question 8 — special-39-03-q08

Source patterns: cbse_sqp_2017-18_electricity_q26
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Identifies the common resistance from several measurements and diagnoses a transcription error.

Rationale tags: evidence-and-reasoning

Question 9 — special-39-03-q09

Source patterns: ncert-11

NCERT references: §11.5, printed p.179: resistance and resistivity vary with temperature; Table 11.2 values at 20 °C

Competency classification: No

Competency marks: 0

Competency rationale: Foundational practical recognition that a tabulated resistivity is temperature-specific; no numerical trend, material classification or experimental diagnosis is required.

Rationale tags: foundational

Question 10 — special-39-03-q10

Source patterns: SQP_2024-25_q25_A
ncert-11

NCERT references: §11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 1

Competency rationale: Must trace a branch switch and distinguish branch current from total source current.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 11 — special-39-03-q11

Source patterns: SQP_2023-24_q32
ncert-11

NCERT references: §11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: This is a routine proportional substitution in $H = I^2Rt$; conservatively classified foundational.

Rationale tags: foundational

Question 12 — special-39-03-q12

Source patterns: SQP_2024-25_q25_B
ncert-11

NCERT references: §11.6.1 / Activities 11.4–11.5, printed pp.182–184
§11.6.2 / Activity 11.6, printed pp.185–186
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Combines equivalent-resistance comparison with equal-energy duration, rather than merely calculating resistance.

Rationale tags: evidence-and-reasoning

Question 13 — special-39-03-q13

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

§11.2, printed p.173

§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Reconciles equal charge-flow rates with a nonzero energy transfer; rejects charge-consumption reasoning.

Rationale tags: evidence-and-reasoning

Question 14 — special-39-03-q14

Source patterns: cbse_sqp_2017-18_electricity_q26
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Diagnoses the sign of an offset and corrects measured voltage before interpreting resistance.

Rationale tags: evidence-and-reasoning

Question 15 — special-39-03-q15

Source patterns: pyq-2018-910a1f81e814c508e7ea459d
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–179

Competency classification: No

Competency marks: 0

Competency rationale: Recall of material properties from the textbook, not a data-based material selection task.

Rationale tags: foundational

Question 16 — special-39-03-q16

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Must read interval boundaries and compare areas as power $\times$ time, including the off interval.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 17 — special-39-03-q17

Source patterns: SQP_2023-24_q32
ncert-11

NCERT references: §11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: The reason supplies $H = V^2t/R$ directly; testing a doubled denominator is foundational scaling.

Rationale tags: foundational

Question 18 — special-39-03-q18

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172
§11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Tests two foundational facts and their lack of explanatory relationship.

Rationale tags: foundational

Question 19 — special-39-03-q19

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172
§11.6.1 / Activities 11.4–11.5, printed pp.182–184

Competency classification: No

Competency marks: 0

Competency rationale: Both the series-current fact and its conservation explanation are supplied; conservatively foundational.

Rationale tags: foundational

Question 20 — special-39-03-q20

Source patterns: pyq-2022-4f106d54aeecf637b6b663fc
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Routine voltage/time scaling plus recognition of the explicitly fixed resistance; foundational.

Rationale tags: foundational

Question 21 — special-39-03-q21

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

Competency classification: Yes

Competency marks: 1

Competency rationale: The first mark is routine $Q = It$; the second tests an unweighted-average misconception using unequal durations.

Rationale tags: evidence-and-reasoning

Question 22 — special-39-03-q22

Source patterns: ncert-11

NCERT references: §11.2, printed p.173

Competency classification: Yes

Competency marks: 1

Competency rationale: Voltage calculations are foundational; the explanatory comparison distinguishes total energy from energy per coulomb.

Rationale tags: evidence-and-reasoning

Question 23 — special-39-03-q23

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

§11.2, printed p.173

§11.6.1 / Activities 11.4–11.5, printed pp.182–184

Competency classification: Yes

Competency marks: 2

Competency rationale: Requires tracing the voltmeter nodes, calculating series current, and distinguishing component voltage from whole-circuit voltage.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 24 — special-39-03-q24

Source patterns: SQP_2023-24_q32
ncert-11

NCERT references: §11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: Standard derivation and the conditions of proportionality are foundational, not automatically competency.

Rationale tags: foundational

Question 25 — special-39-03-q25

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Part (i) compares complete equal-outcome schedules. Part (ii) is a routine cost multiplication and receives no competency credit.

Rationale tags: evidence-and-reasoning

Question 26 — special-39-03-q26

Source patterns: cbse_sqp_2018-19_electricity_q13
pyq-2018-910a1f81e814c508e7ea459d
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–179
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 2

Competency rationale: Detects length as a confounder and designs a comparison that isolates material.

Rationale tags: evidence-and-reasoning

Question 27 — special-39-03-q27

Source patterns: cbse_sqp_2018-19_electricity_q13
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–179
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2

Competency rationale: Part (i) is routine; part (ii) transfers a diameter change through area and resistance into a voltage-controlled energy decision.

Rationale tags: evidence-and-reasoning

Question 28 — special-39-03-q28

Source patterns: SQP_2023-24_q32
ncert-11

NCERT references: §11.7, printed pp.188–189: $H = VIt$, $H = I^2Rt$ and current recalculation at known voltage
§11.1, printed p.172: charge and current
§11.5, printed p.179: resistance can vary with temperature

Competency classification: Yes

Competency marks: 2

Competency rationale: Part (i) is routine stagewise substitution (0). Part (ii) diagnoses why multiplying separately averaged current and resistance misrepresents their paired stages (1). Part (iii) distinguishes a valid fixed-voltage average-current method using charge or additive energy, rather than rejecting all averages (1).

Rationale tags: piecewise-energy
method-comparison
error-diagnosis

Question 29 — special-39-03-q29

Source patterns: pyq-2022-4f106d54aeecf637b6b663fc
ncert-11

NCERT references: §11.8, printed pp.191–192
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 2

Competency rationale: The resistance-from-rating step is foundational; comparing both altered-voltage performances against a deadline earns two competency marks.

Rationale tags: evidence-and-reasoning

Question 30 — special-39-03-q30

Source patterns: SQP_2023-24_q32
pyq-2023-246eac3017f0793c27bba6a6
ncert-11

NCERT references: §11.7, printed pp.188–189
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2

Competency rationale: Reading the heat ratio in (i) is foundational. The two marks in (ii) interpret heating rates to reject equal current and identify the common voltage.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 31 — special-39-03-q31

Source patterns: SQP_2024-25_q25_A
ncert-11

NCERT references: §11.6.1 / Activities 11.4–11.5, printed pp.182–184
§11.6.2 / Activity 11.6, printed pp.185–186
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 2

Competency rationale: The pre-fault baseline circuit calculation is foundational. Reconstructing the faulted circuit and explaining increased surviving-branch heat earn two competency marks.

Rationale tags: evidence-and-reasoning

Question 32 — special-39-03-q32

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Distinguishes a peak constraint from a time-accumulated constraint and selects a feasible schedule.

Rationale tags: evidence-and-reasoning

Question 33 — special-39-03-q33

Source patterns: ncert-11

NCERT references: §11.1, printed p.172: ammeter measures current and is connected in series
§11.4 / Activity 11.1, printed pp.175–176: V/I measurements, resistance and straight-line relation at constant temperature

Competency classification: Yes

Competency marks: 2

Competency rationale: Part (i) is a routine V/I calculation (0). Part (ii) uses independent same-condition meter-swap evidence to discriminate a changed component from an instrument scale error (1). Part (iii) explains why a multiplicative error passes both zero and straight-line checks (1). No additive offset, contact resistance, heating anomaly or voltmeter-span error is involved.

Rationale tags: experimental-inference
meter-swap-evidence
scale-factor-error

Question 34 — special-39-03-q34

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2.5

Competency rationale: Part (i) is routine accumulation of specified power-time blocks (0). Part (ii) tests both complete daily schedules against a cap (2). In part (iii), cost multiplication is foundational (0), but explaining why peak power alone is insufficient earns 0.5.

Rationale tags: evidence-and-reasoning

Question 35 — special-39-03-q35

Source patterns: pyq-2022-4f106d54aeecf637b6b663fc
sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 2

Competency rationale: Resistance, altered-voltage power, and inversion of the explicitly supplied useful fraction are foundational. The deadline decision earns 1 and the quantitative diagnosis of the misused rating earns 1.

Rationale tags: evidence-and-reasoning

Question 36 — special-39-03-q36

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 5

Competency rationale: Requires decomposing a supplied graph, testing independent limits and locating a cumulative-energy threshold inside an interval. No slope differentiation or integration is used.

Rationale tags: evidence-and-reasoning
diagram-dependent

Question 37 — special-39-03-q37

Source patterns: ncert-11
sqp_2021-22_term2_q11

NCERT references: §11.1, printed pp.171–172
§11.2, printed p.173
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Part (i) is routine. The remaining marks require a whole-cycle decision and a distinction between deliverable charge-throughput and charge destruction.

Rationale tags: evidence-and-reasoning

Question 38 — special-39-03-q38

Source patterns: SQP_2023-24_q32
ncert-11

NCERT references: §11.7, printed pp.188–189
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 3

Competency rationale: Part (i) is routine (0). Part (ii) diagnoses a reversed data log against fixed-voltage predictions (1); part (iii) qualifies an invalid universal claim (1); part (iv) identifies a time confounder and designs a current-only repeat (1).

Rationale tags: evidence-and-reasoning

Question 39 — special-39-03-q39

Source patterns: cbse_sqp_2018-19_electricity_q13
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–179
§11.7, printed pp.188–189
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2

Competency rationale: Required resistance and three direct geometry substitutions are foundational. Selecting all compliant candidates with a heat check earns 1; explaining cancellation of length and area effects earns 1.

Rationale tags: evidence-and-reasoning

Source reference details

id: SQP_2023-24_q32

title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 32

session: 2023-24

exam_type: SQP

original_section: Section-C

language: en

pages: 5

page: 5

bank_verified: Yes

bank_answer_verified: No

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: SQP_2024-25_q25_A

title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 25 A

session: 2024-25

exam_type: SQP

original_section: Section-B

language: en

pages: 6

page: 6

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: SQP_2024-25_q25_B

title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 25 B

session: 2024-25

exam_type: SQP

original_section: Section-B

language: en

pages: 6

7

page: 6

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2017-18_electricity_q26

title: Sample Question Paper 2017-18 — Science — Class - X

url: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 26

session: 2017-18

exam_type: SQP

original_section: SECTION - B

language: en

pages: 3

4

page: 3

bank_verified: Yes

bank_answer_verified: No

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2018-19_electricity_q13

title: Class-X Science-086 SAMPLE QUESTION PAPER-19

url: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 13

session: 2018-19

exam_type: SQP

original_section: SECTION C

language: en

pages: 3

page: 3

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: ncert-11

title: NCERT Science, Class X, Chapter 11: Electricity

url: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

source_role: conceptual-scope

local_source_record: research/sqp/archive/jesc111.txt

notes: Conceptual scope and activity reference only; not provenance of original questions. No question-bank verification flags are inferred.

id: pyq-2018-910a1f81e814c508e7ea459d

title: CBSE — Class X Science — 2018 — 31/1 — main

url: https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip

source_role: design-pattern

local_source_record: research/pyq/question_bank.json

question_number: 13

year: 2018

session: main

exam_type: PYQ

paper_code: 31/1

language: English

source_title_generated: Yes
session_inferred: Yes
original_section: Section - A
pages: 6
page: 6
bank_verified: Yes
bank_answer_verified: No
bank_official_source_verified: Yes
bank_text_verified: Yes
bank_visual_verified: Yes
bank_year_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2022-4f106d54aeecf637b6b663fc
title: CBSE — Class X Science — 2022 — compartment — 31/6/1
url: <https://www.cbse.gov.in/cbsenew/question-paper/2022-COMPTT/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 12
year: 2022
session: compartment
exam_type: PYQ
paper_code: 31/6/1
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 11
page: 11
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2023-246eac3017f0793c27bba6a6
title: CBSE — Class X Science — 2023 — main — 31/2/1
url: <https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 25
year: 2023
session: main
exam_type: PYQ
paper_code: 31/2/1
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 15
page: 15
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: sqp_2021-22_term2_q11
title: CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2
url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf
source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 11

session: 2021-22

exam_type: SQP

original_section: Section B

term: 2

language: en

pages: 4

page: 4

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.